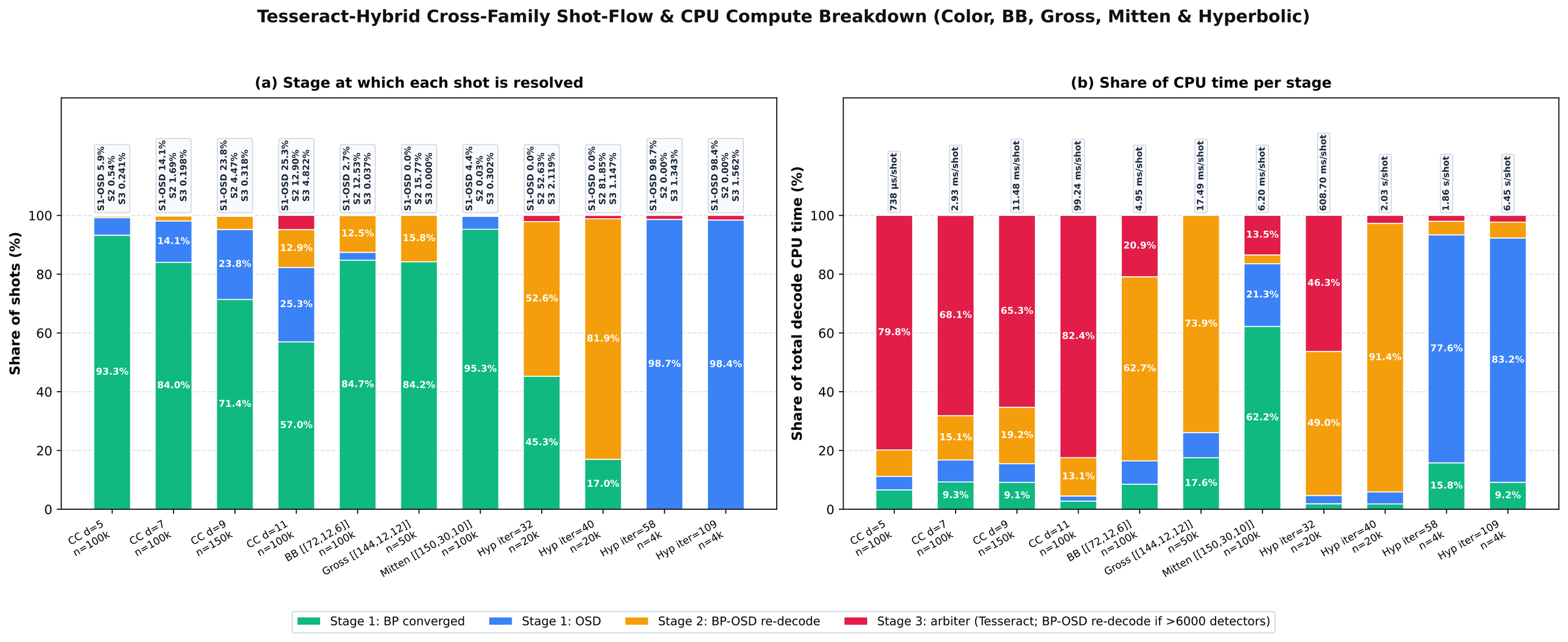
Tesseract-Hybrid Cross-Family Shot-Flow & CPU Compute Breakdown (Color, BB, Gross, Mitten & Hyperbolic)
(a) Stage at which each shot is resolved
Share of shots (%)
S1-OSD 5.9% S2 0.54% S3 0.241%
S1-OSD 14.1% S2 1.69% S3 0.198%
S1-OSD 23.8% S2 4.47% S3 0.318%
S1-OSD 25.3% S2 12.90% S3 4.822%
S1-OSD 2.7% S2 12.53% S3 0.037%
S1-OSD 0.0% S2 15.77% S3 0.000%
S1-OSD 4.4% S2 0.03% S3 0.302%
S1-OSD 0.0% S2 52.63% S3 2.119%
S1-OSD 0.0% S2 81.85% S3 1.147%
S1-OSD 98.7% S2 0.00% S3 1.343%
S1-OSD 98.4% S2 0.00% S3 1.562%
93.3%
84.0%
14.1%
71.4%
23.8%
57.0%
25.3%
12.9%
84.7%
12.5%
84.2%
15.8%
95.3%
45.3%
52.6%
17.0%
81.9%
98.7%
98.4%
CC d=5 n=100k
CC d=7 n=100k
CC d=9 n=150k
CC d=11 n=100k
BB [[72,12,6]] n=100k
Gross [[144,12,12]] n=50k
Mitten [[150,30,10]] n=100k
Hyp iter=32 n=20k
Hyp iter=40 n=20k
Hyp iter=58 n=4k
Hyp iter=109 n=4k
(b) Share of CPU time per stage
Share of total decode CPU time (%)
738 µs/shot
2.93 ms/shot
11.48 ms/shot
99.24 ms/shot
4.95 ms/shot
17.49 ms/shot
6.20 ms/shot
608.70 ms/shot
2.03 s/shot
1.86 s/shot
6.45 s/shot
79.8%
68.1%
15.1%
9.3%
65.3%
19.2%
9.1%
82.4%
13.1%
20.9%
62.7%
73.9%
17.6%
13.5%
21.3%
62.2%
46.3%
49.0%
91.4%
77.6%
15.8%
83.2%
9.2%
Stage 1: BP converged
Stage 1: OSD
Stage 2: BP-OSD re-decode
Stage 3: arbiter (Tesseract; BP-OSD re-decode if >6000 detectors)